Stealing Speed from Planets: Measuring a Gravity Assist Using Real NASA Spacecraft Data

Abstract

Gravity assists allow spacecraft to change velocity without expending propellant, yet the effect is usually described in textbook terms rather than measured. In this work, the velocity gained by Voyager 2 (1979) and New Horizons (2007) during their Jupiter flybys is measured directly from reconstructed trajectories in the NASA JPL Horizons system. After correcting for the deceleration caused by solar gravity, Jupiter is found to have increased the heliocentric speed of Voyager 2 by 10.60 ± 0.06 km/s and that of New Horizons by 3.20 ± 0.03 km/s. The observed velocity changes, 11.95 ± 0.23 km/s and 5.14 ± 0.01 km/s, agree with the patched-conic prediction to within 4.3% and 0.6% respectively, and the observed deflection angles match theory to within 0.6°. A numerical simulation independently reproduces the predicted deflections to within 0.05°. Applying the Tsiolkovsky rocket equation, an equivalent propulsive manoeuvre with the spacecraft's own hydrazine thrusters would have required approximately 97 t of propellant for Voyager 2 and 1.6 t for New Horizons. The larger discrepancy for Voyager 2 is attributed to its slower encounter and to measurement end points lying inside Jupiter's sphere of influence.

Keywords: gravity assist, patched conics, Voyager 2, New Horizons, JPL Horizons, rocket equation

1. Introduction

Reaching the outer Solar System requires large changes in velocity. Chemical propulsion alone scales poorly to this task: the rocket equation makes the required propellant mass grow exponentially with the velocity change, so each additional kilometre per second is more expensive than the last. Most deep-space missions therefore rely on gravity assists, in which a close pass of a planet redirects the spacecraft and alters its speed relative to the Sun.

Two missions illustrate the technique particularly well. Voyager 2 passed Jupiter on 9 July 1979 on its way to Saturn, Uranus and Neptune, and New Horizons passed Jupiter on 28 February 2007 en route to Pluto. Their trajectories are publicly available through the JPL Horizons ephemeris service, which makes it possible to measure the effect of each flyby rather than simply assume it.

This paper addresses the following question: *can the speed gained by a spacecraft during a planetary flyby be measured from real trajectory data, and does it match gravity-assist physics?* To answer it, we (i) measure the heliocentric velocity change at Jupiter for both spacecraft, (ii) compare the result with the patched-conic model, (iii) verify the model with an independent numerical simulation, and (iv) estimate the propellant that an equivalent rocket manoeuvre would have required.

The remainder of the paper is organised as follows. Section 2 summarises the relevant theory. Section 3 describes the data and the measurement method. Section 4 presents the results, Section 5 the fuel estimate, and Section 6 discusses uncertainties and limitations. Section 7 concludes.

2. Theory

2.1 Reference frames

In a frame moving with the planet, a flyby is an encounter with a conservative central force. The spacecraft approaches on a hyperbolic trajectory and departs with the same speed far from the planet, the hyperbolic excess speed v∞. Only its direction changes, through a deflection angle δ. In the heliocentric frame the spacecraft velocity is the vector sum of the planet's orbital velocity and the planet-relative velocity:

$$\vec{v}_{\odot} = \vec{V}_J + \vec{v}_{\infty}$$

Because the incoming and outgoing v∞ vectors point in different directions, adding the same planetary velocity V_J (about 13 km/s for Jupiter) to each produces heliocentric velocities of different magnitude. The energy gained by the spacecraft is balanced by an immeasurably small loss in Jupiter's orbital energy.

2.2 Hyperbolic flyby

For a two-body hyperbola with periapsis distance r_p, the deflection angle is

$$\sin\left(\frac{\delta}{2}\right) = \frac{1}{1 + r_p v_{\infty}^2/\mu_J}$$

where $\mu_J = 1.267 \times 10^8$ km³ s⁻² is Jupiter's gravitational parameter. The magnitude of the velocity change, which is identical in both frames, follows directly:

$$|\Delta\vec{v}| = 2v_{\infty}\sin\left(\frac{\delta}{2}\right)$$

A closer pass or a slower approach therefore produces a larger deflection. For a fixed v∞ the velocity change can never exceed 2v∞, which corresponds to a complete reversal.

2.3 Sphere of influence

The patched-conic approximation treats the flyby as a pure two-body problem inside the planet's sphere of influence (SOI) and as heliocentric motion outside it. The SOI radius is

$$r_{SOI} = a_J\left(\frac{\mu_J}{\mu_{\odot}}\right)^{2/5}$$

which is approximately 49×10^6 km for Jupiter.

2.4 Rocket equation

To express the flyby as a propellant saving, the Tsiolkovsky rocket equation is used:

$$\Delta v = I_{sp}\, g_0 \ln\left(\frac{m_0}{m_f}\right) \quad \Rightarrow \quad m_{prop} = m_f\left(e^{\Delta v/(I_{sp}g_0)} - 1\right)$$

where I_{sp} is the specific impulse of the engine, $g_0 = 9.80665$ m s⁻², and m_f is the spacecraft mass after the burn.

3. Data and Methods

3.1 Trajectory data

State vectors (position and velocity) were retrieved from the JPL Horizons system using the `astroquery` Python package. For each spacecraft, data were obtained at hourly intervals over a window of ±60 days centred on closest approach, in two frames: Sun-centred (Horizons centre code 500@10) and Jupiter-centred (500@599). Jupiter's heliocentric state was retrieved over the same intervals. All vectors are referred to the ecliptic plane and were converted from AU and AU/day to km and km/s. Table 1 summarises the data set.

Table 1. Spacecraft and data windows.

Spacecraft	Horizons ID	Closest approach (UTC)	Data window	Samples per frame
Voyager 2	-32	1979-07-09 22:00	1979-05-10 to 1979-09-07	2,881
New Horizons	-98	2007-02-28 06:00	2006-12-30 to 2007-04-29	2,881

Because hourly sampling can miss the true periapsis, a second query at one-minute resolution was made over ±2 hours around each closest approach to determine r_p.

[1]:
```
import numpy as np, pandas as pd, matplotlib.pyplot as plt
from astroquery.jplhorizons import Horizons
from astropy.time import Time

jd = Time('2007-02-28').jd          # converts the date to a Julian Date
obj = Horizons(id='599', location='500@10', epochs=jd)
v = obj.vectors()
print(v['targetname'][0])
print("Distance from Sun (AU):", float(np.sqrt(v['x']**2 + v['y']**2 + v['z']**2)[0]))
```

```
Jupiter (599)
Distance from Sun (AU): 5.349396601537589
```

[2]:
```
import numpy as np, pandas as pd
from astroquery.jplhorizons import Horizons

AU  = 149597870.7   # km
DAY = 86400.0       # s

def get_vectors(target, center, start, stop, step='1h'):
    obj = Horizons(id=target, location=center,
                   epochs={'start': start, 'stop': stop, 'step': step})
    t = obj.vectors(refplane='ecliptic').to_pandas()
    df = pd.DataFrame({
        'jd':   t['datetime_jd'],
        'date': t['datetime_str'],
        'x':  t['x']*AU,  'y':  t['y']*AU,  'z':  t['z']*AU,          # km
        'vx': t['vx']*AU/DAY, 'vy': t['vy']*AU/DAY, 'vz': t['vz']*AU/DAY,  # km/s
    })
    df['r']     = np.sqrt(df.x**2 + df.y**2 + df.z**2)
    df['speed'] = np.sqrt(df.vx**2 + df.vy**2 + df.vz**2)
    return df
```

[3]:
```
flybys = {
    'voyager2':     {'id': '-32', 'start': '1979-05-10', 'stop': '1979-09-07'},
    'newhorizons':  {'id': '-98', 'start': '2006-12-30', 'stop': '2007-04-29'},
}

frames = {'sun': '500@10', 'jup': '500@599'}

for name, f in flybys.items():
    for fname, center in frames.items():
        df = get_vectors(f['id'], center, f['start'], f['stop'])
        df.to_csv(f'data/{name}_{fname}.csv', index=False)
        print(f'{name}_{fname}: {len(df)} rows')

    # Jupiter around the Sun, same dates (for plots)
    jup = get_vectors('599', '500@10', f['start'], f['stop'])
    jup.to_csv(f'data/jupiter_{name}.csv', index=False)
    print(f'jupiter_{name}: {len(jup)} rows')
```

```
voyager2_sun: 2881 rows
voyager2_jup: 2881 rows
jupiter_voyager2: 2881 rows
newhorizons_sun: 2881 rows
newhorizons_jup: 2881 rows
jupiter_newhorizons: 2881 rows
```

```
[4]: for name in flybys:
         j = pd.read_csv(f'data/{name}_jup.csv')
         i = j['r'].idxmin()
         print(f"{name}: closest approach {j.loc[i,'date']}, "
               f"distance {j.loc[i,'r']:,.0f} km, speed {j.loc[i,'speed']:.1f} km/s")
```

```
voyager2: closest approach A.D. 1979-Jul-09 22:00:00.0000, distance 721,895 km, speed 20.2 km/s
newhorizons: closest approach A.D. 2007-Feb-28 06:00:00.0000, distance 2,304,577 km, speed 21.2 km/s
```

3.2 Measuring the velocity change

A naive estimate of the gain is the difference in heliocentric speed between the start and end of the window. This estimate is biased, because solar gravity decelerates an outbound spacecraft throughout the window. To isolate Jupiter's contribution, the observed heliocentric state at −60 days was propagated to +60 days under solar gravity alone, using an eighth-order Runge–Kutta integrator (DOP853, relative tolerance 10^{-11}). The difference between the observed and propagated velocities at +60 days gives the velocity change attributable to Jupiter. Both its magnitude $|\Delta v|$ and the corresponding change in speed are reported.

3.3 Comparison with theory

From the Jupiter-centred data, v_∞ was computed at both ends of the window using energy conservation, which removes the residual effect of Jupiter's potential at finite distance:

$$v_\infty = \sqrt{v^2 - 2\mu_J / r}$$

The mean of the incoming and outgoing values was used together with r_p to predict δ and $|\Delta v|$ from Section 2.2. The observed deflection was taken as the angle between the planet-relative velocity vectors at −60 and +60 days.

3.4 Numerical simulation

As an independent check on the analytic model, planar flybys were integrated numerically under Jupiter's gravity alone for v_∞ = 7.6 km/s and six periapsis distances between 2 and 80 Jupiter radii (R_J = 71,492 km). Each trajectory was started at 2,000 R_J with an impact parameter chosen to produce the desired r_p and integrated until it returned to the same distance.

3.5 Fuel estimate

The propellant equivalent of each flyby was computed from Section 2.4 using launch masses of 722 kg (Voyager 2) and 478 kg (New Horizons). Three specific impulses were considered: 220 s, representative of the hydrazine thrusters carried by both spacecraft; 320 s, typical of a bipropellant engine; and 450 s, close to the practical limit for chemical propulsion.

4. Results

4.1 Trajectories

The closest approaches recovered from the data, 721,375 km (10.09 R_J) for Voyager 2 and 2,304,505 km (32.23 R_J) for New Horizons, agree with the published encounter geometries. At heliocentric scale both paths appear nearly straight; only Voyager 2 shows a visible change of direction at Jupiter (Figure 1). In the Jupiter-centred frame both trajectories are clear hyperbolas, with Voyager 2 deflected far more strongly (Figure 2).

[5]:
```python
import numpy as np, pandas as pd, matplotlib.pyplot as plt

AU = 149597870.7   # km
RJ = 71492.0       # Jupiter radius, km
names = {'voyager2': 'Voyager 2', 'newhorizons': 'New Horizons'}

data = {}
for n in names:
    data[n] = {'sun': pd.read_csv(f'data/{n}_sun.csv'),
               'jup': pd.read_csv(f'data/{n}_jup.csv'),
               'jupiter': pd.read_csv(f'data/jupiter_{n}.csv')}
    j = data[n]['jup']
    t0 = j.loc[j['r'].idxmin(), 'jd']          # time of closest approach
    for k in data[n]:
        data[n][k]['days'] = data[n][k]['jd'] - t0
```

[6]:
```python
fig, axes = plt.subplots(1, 2, figsize=(13, 6))
for ax, (n, label) in zip(axes, names.items()):
    s, jp = data[n]['sun'], data[n]['jupiter']
    ax.plot(0, 0, '*', color='gold', ms=18, label='Sun')
    ax.plot(jp.x/AU, jp.y/AU, color='orange', lw=3, label='Jupiter')
    ax.plot(s.x/AU, s.y/AU, color='tab:blue', label=label)
    i = s['days'].abs().idxmin()
    ax.plot(s.x[i]/AU, s.y[i]/AU, 'ko', label='Closest approach')
    ax.set_aspect('equal'); ax.grid(alpha=0.3)
    ax.set_xlabel('x (AU)'); ax.set_ylabel('y (AU)')
    ax.set_title(f'{label}: Sun frame (±60 days)'); ax.legend()
plt.tight_layout(); plt.savefig('plots/1_sun_frame.png', dpi=200); plt.show()
```

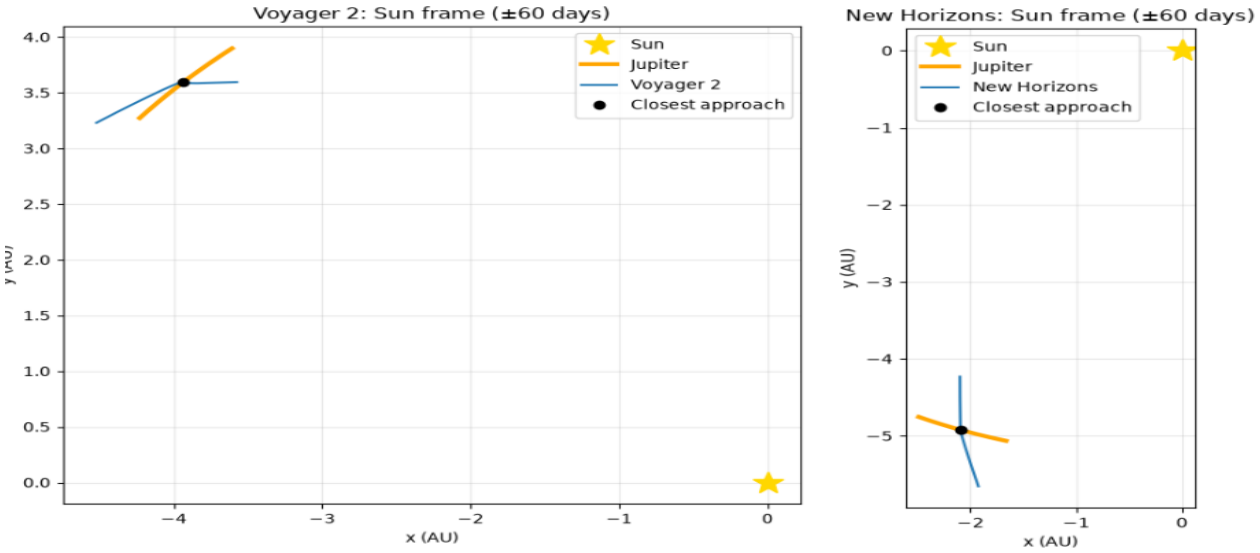


Figure 1. Heliocentric trajectories of Voyager 2 (left) and New Horizons (right) over ±60 days, projected on the ecliptic plane, with Jupiter's orbit and the point of closest approach.

```
[7]: fig, axes = plt.subplots(1, 2, figsize=(13, 6))
     for ax, (n, label) in zip(axes, names.items()):
         j = data[n]['jup']
         ax.add_patch(plt.Circle((0, 0), 1, color='orange'))   # Jupiter to scale
         ax.plot(j.x/RJ, j.y/RJ, color='tab:blue', label=label)
         i = j['r'].idxmin()
         ax.plot(j.x[i]/RJ, j.y[i]/RJ, 'ko', label=f"Closest: {j.r[i]/RJ:.1f} RJ")
         ax.set_xlim(-100, 100); ax.set_ylim(-100, 100)
         ax.set_aspect('equal'); ax.grid(alpha=0.3)
         ax.set_xlabel('x (Jupiter radii)'); ax.set_ylabel('y (Jupiter radii)')
         ax.set_title(f'{label}: Jupiter frame'); ax.legend()
     plt.tight_layout(); plt.savefig('plots/2_jupiter_frame.png', dpi=200); plt.show()
```

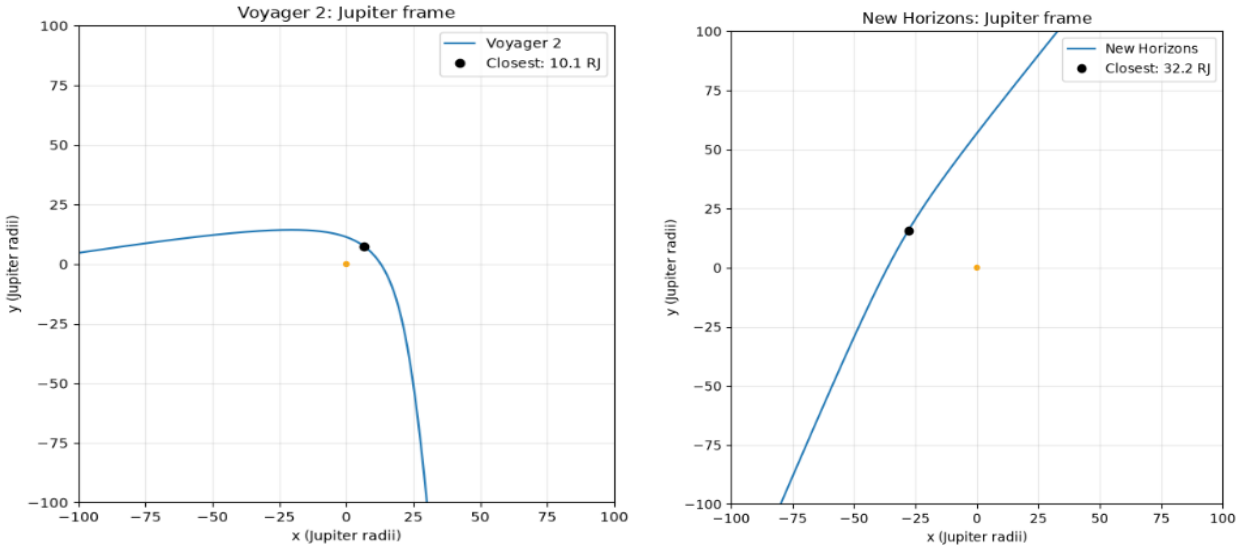


Figure 2. Jupiter-centred trajectories, in units of Jupiter radii.

The speed profiles in Figure 3 show the signature of a gravity assist. Relative to Jupiter, each spacecraft's speed rises to a peak at periapsis and returns to its approach value, about 8 km/s for Voyager 2 and 18.6 km/s for New Horizons. Relative to the Sun, the post-encounter speed settles at a higher level than before: roughly 10 km/s rising to 19.5 km/s for Voyager 2. Outside the encounter the heliocentric speed of New Horizons declines steadily, reflecting the solar deceleration discussed in Section 3.2.

```
[8]: fig, axes = plt.subplots(2, 1, figsize=(11, 8), sharex=True)
     for ax, (n, label) in zip(axes, names.items()):
         ax.plot(data[n]['sun'].days, data[n]['sun'].speed, label='Relative to Sun')
         ax.plot(data[n]['jup'].days, data[n]['jup'].speed, label='Relative to Jupiter')
         ax.axvline(0, color='gray', ls='--', lw=1)
         ax.set_ylabel('Speed (km/s)'); ax.set_title(label)
         ax.grid(alpha=0.3); ax.legend()
     axes[-1].set_xlabel('Days from closest approach')
     plt.tight_layout(); plt.savefig('plots/3_speed_vs_time.png', dpi=200); plt.show()
```

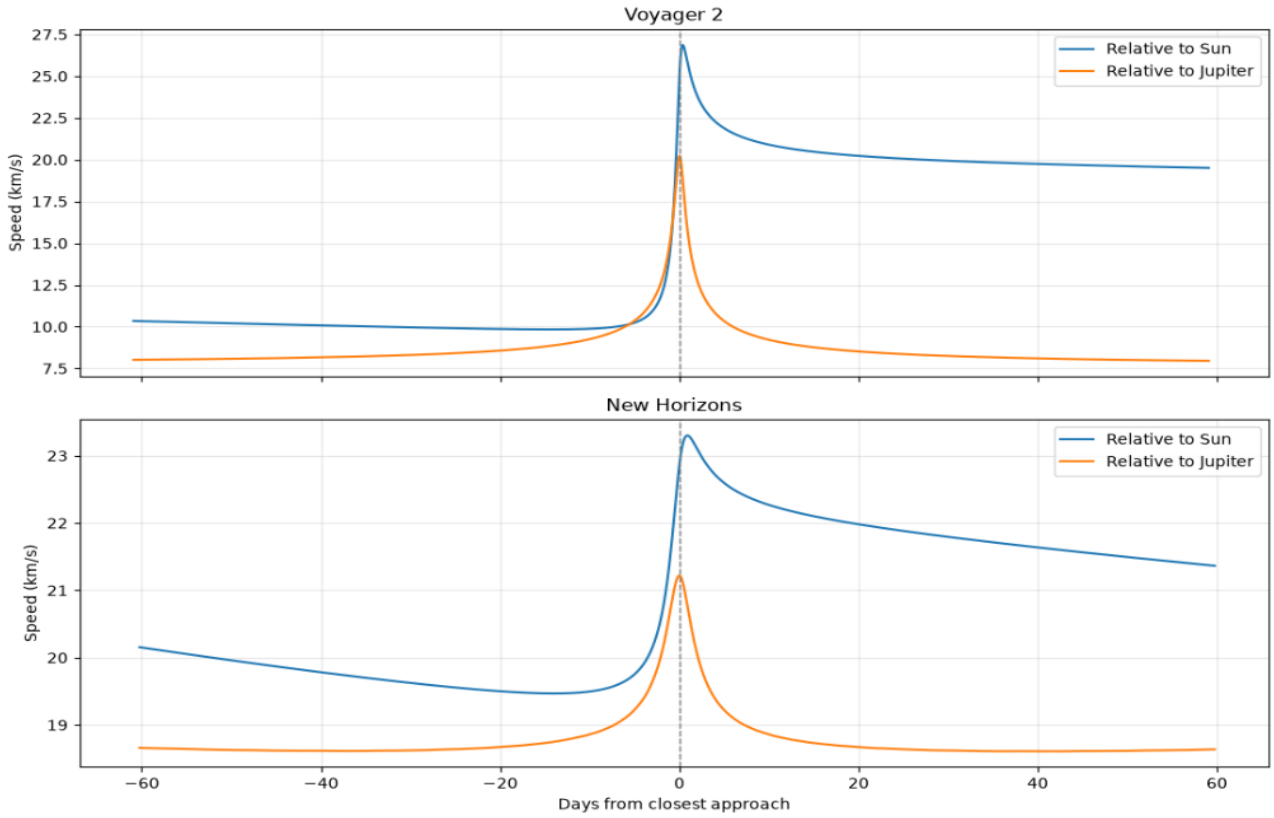


Figure 3. Speed relative to the Sun and to Jupiter as a function of time from closest approach.

4.2 Measured velocity change

Table 2 lists the measured quantities. Had Jupiter not been present, Voyager 2 would have been travelling at 8.91 km/s at the end of the window instead of 19.51 km/s. For New Horizons the naive estimate of 1.21 km/s understates the solar-corrected gain of 3.20 km/s by almost a factor of three, demonstrating that the correction is essential. In both cases the speed gain is smaller than |Δv|, since part of the velocity change acts to rotate the velocity vector rather than to lengthen it.

[9]:
```
MU_SUN = 1.32712440018e11   # km^3/s^2
MU_JUP = 1.26686534e8       # km^3/s^2

for n, label in names.items():
    j, jp = data[n]['jup'], data[n]['jupiter']
    r_soi = jp['r'].mean() * (MU_JUP/MU_SUN)**0.4
    print(f"{label}: SOI radius {r_soi/1e6:.0f} million km | "
          f"distance from Jupiter at -60 d: {j.r.iloc[0]/1e6:.0f}, "
          f"at +60 d: {j.r.iloc[-1]/1e6:.0f} million km")
```

```
Voyager 2: SOI radius 49 million km | distance from Jupiter at -60 d: 46, at +60 d: 44 million km
New Horizons: SOI radius 50 million km | distance from Jupiter at -60 d: 98, at +60 d: 97 million km
```

```
[10]: from scipy.integrate import solve_ivp

def sun_only(t, s):
    r = s[:3]
    return np.concatenate([s[3:], -MU_SUN * r / np.linalg.norm(r)**3])

results = {}
for n, label in names.items():
    s = data[n]['sun']
    b, a = s.iloc[0], s.iloc[-1]                    # before (-60 d), after (+60 d)
    r_b = b[['x','y','z']].to_numpy(float)
    v_b = b[['vx','vy','vz']].to_numpy(float)
    v_a = a[['vx','vy','vz']].to_numpy(float)

    T = (a.jd - b.jd) * 86400                       # seconds
    sol = solve_ivp(sun_only, [0, T], np.concatenate([r_b, v_b]),
                    method='DOP853', rtol=1e-11, atol=1e-6)
    v_nojup = sol.y[3:, -1]

    naive     = np.linalg.norm(v_a) - np.linalg.norm(v_b)
    corrected = np.linalg.norm(v_a) - np.linalg.norm(v_nojup)
    dv_vec    = np.linalg.norm(v_a - v_nojup)
    results[n] = {'naive': naive, 'speed_gain': corrected, 'dv_obs': dv_vec}

    print(f"{label}")
    print(f"  Speed before (-60 d):          {np.linalg.norm(v_b):6.2f} km/s")
    print(f"  Speed after  (+60 d):          {np.linalg.norm(v_a):6.2f} km/s")
    print(f"  No-Jupiter speed at +60 d:     {np.linalg.norm(v_nojup):6.2f} km/s")
    print(f"  Naive gain:                    {naive:6.2f} km/s")
    print(f"  Corrected speed gain:          {corrected:6.2f} km/s")
    print(f"  Velocity change |Δv| (vector): {dv_vec:6.2f} km/s\n")
```

```
Voyager 2
  Speed before (-60 d):          10.35 km/s
  Speed after  (+60 d):          19.51 km/s
  No-Jupiter speed at +60 d:      8.91 km/s
  Naive gain:                     9.16 km/s
  Corrected speed gain:          10.60 km/s
  Velocity change |Δv| (vector):  11.95 km/s

New Horizons
  Speed before (-60 d):          20.16 km/s
  Speed after  (+60 d):          21.36 km/s
  No-Jupiter speed at +60 d:     18.17 km/s
  Naive gain:                     1.21 km/s
  Corrected speed gain:           3.20 km/s
  Velocity change |Δv| (vector):   5.14 km/s
```

Table 2. Heliocentric speeds and velocity change at Jupiter (±60-day window).

4.3 Comparison with the patched-conic model

Table 3 compares the observations with the predictions of Section 2.2. The incoming and outgoing values of v_∞ agree to within 1.0% for Voyager 2 and 0.1% for New Horizons, consistent with energy conservation in the planet's frame. The predicted deflection angles match the observed ones to within 0.6°, and the predicted $|\Delta v|$ agrees with the measurement to within 4.3% for Voyager 2 and 0.6% for New Horizons.

[11]:
```python
from astropy.time import Time

def vinf(row):
    return np.sqrt(row['speed']**2 - 2*MU_JUP/row['r'])

def angle_deg(a, b):
    return np.degrees(np.arccos(np.dot(a, b) / (np.linalg.norm(a)*np.linalg.norm(b))))

rows = {}
for n, label in names.items():
    j = data[n]['jup']

    # precise closest approach (1-minute data, ±2 hours)
    t_ca  = j.loc[j['r'].idxmin(), 'jd']
    start = Time(t_ca - 2/24, format='jd').strftime('%Y-%m-%d %H:%M')
    stop  = Time(t_ca + 2/24, format='jd').strftime('%Y-%m-%d %H:%M')
    sc_id = {'voyager2': '-32', 'newhorizons': '-98'}[n]
    fine  = get_vectors(sc_id, '500@599', start, stop, step='1m')
    r_p   = fine['r'].min()

    # v-infinity in and out
    first, last = j.iloc[0], j.iloc[-1]
    v_in, v_out = vinf(first), vinf(last)
    v_inf = (v_in + v_out) / 2

    # observed bend angle
    d_obs = angle_deg(first[['vx','vy','vz']].to_numpy(float),
                      last[['vx','vy','vz']].to_numpy(float))

    # theory
    s = 1 / (1 + r_p * v_inf**2 / MU_JUP)
    d_th  = 2 * np.degrees(np.arcsin(s))
    dv_th = 2 * v_inf * s
    dv_obs = results[n]['dv_obs']
    results[n].update({'v_inf': v_inf, 'r_p': r_p, 'dv_th': dv_th})

    rows[label] = {
        'v∞ in (km/s)': v_in, 'v∞ out (km/s)': v_out,
        'r_p (km)': r_p, 'r_p (RJ)': r_p/RJ,
        'Bend δ observed (°)': d_obs, 'Bend δ theory (°)': d_th,
        '|Δv| observed (km/s)': dv_obs, '|Δv| theory (km/s)': dv_th,
        'Difference (%)': 100*(dv_obs - dv_th)/dv_th,
    }

table = pd.DataFrame(rows)
table.to_csv('data/theory_vs_observed.csv')
table.round(2)
```

[11]:

	Voyager 2	New Horizons
v∞ in (km/s)	7.67	18.59
v∞ out (km/s)	7.59	18.57
r_p (km)	721375.48	2304504.78
r_p (RJ)	10.09	32.23
Bend δ observed (°)	96.82	15.88
Bend δ theory (°)	97.37	15.79
|Δv| observed (km/s)	11.95	5.14
|Δv| theory (km/s)	11.46	5.10
Difference (%)	4.28	0.63

Table 3. Observed and predicted encounter parameters.

4.4 Numerical simulation

The simulated deflections agree with the analytic formula to within 0.05° at every periapsis distance tested (Table 4, Figure 4). The results also show diminishing returns: reducing r_p from 80 to 10 R_J nearly triples $|\Delta v|$, whereas a further reduction to 2 R_J adds only 2.8 km/s, as the velocity change approaches its limit of $2v_\infty$ = 15.2 km/s. At 10 R_J the simulation gives 11.46 km/s, identical to the analytic prediction for Voyager 2.

[13]:
```python
def jup_only(t, s):
    r = s[:2]
    return np.concatenate([s[2:], -MU_JUP * r / np.linalg.norm(r)**3])

def simulate(r_p, v_inf, R0=2000*RJ):
    b  = r_p * np.sqrt(1 + 2*MU_JUP/(r_p * v_inf**2))   # aiming distance
    v0 = np.sqrt(v_inf**2 + 2*MU_JUP/R0)                # speed at start
    y0 = b * v_inf / v0
    s0 = [-np.sqrt(R0**2 - y0**2), y0, v0, 0.0]

    leave = lambda t, s: np.hypot(s[0], s[1]) - R0
    leave.terminal, leave.direction = True, 1
    sol = solve_ivp(jup_only, [0, 10*R0/v_inf], s0, events=leave,
                    method='DOP853', rtol=1e-10, atol=1e-3)

    v_start, v_end = sol.y[2:, 0], sol.y[2:, -1]
    delta = angle_deg(v_start, v_end)
    dv = 2 * v_inf * np.sin(np.radians(delta)/2)
    return sol, delta, dv

v_test = 7.6                          # Voyager 2-like approach speed, km/s
fig, ax = plt.subplots(figsize=(8, 8))
ax.add_patch(plt.Circle((0, 0), 1, color='orange'))
print(f"v∞ = {v_test} km/s")
for rp_RJ in [2, 5, 10, 20, 40, 80]:
    sol, d, dv = simulate(rp_RJ*RJ, v_test)
    d_th = 2*np.degrees(np.arcsin(1/(1 + rp_RJ*RJ*v_test**2/MU_JUP)))
    ax.plot(sol.y[0]/RJ, sol.y[1]/RJ, label=f'r_p = {rp_RJ} RJ')
    print(f"  r_p = {rp_RJ:3d} RJ: bend simulated {d:6.2f}°, theory {d_th:6.2f}°, Δv = {dv:5.2f} km/s")
ax.set_xlim(-100, 100); ax.set_ylim(-100, 100); ax.set_aspect('equal')
ax.set_xlabel('x (Jupiter radii)'); ax.set_ylabel('y (Jupiter radii)')
ax.set_title(f'Simulated flybys, v∞ = {v_test} km/s'); ax.grid(alpha=0.3); ax.legend()
plt.savefig('plots/4_simulated_flybys.png', dpi=200); plt.show()
```

```
v∞ = 7.6 km/s
  r_p =   2 RJ: bend simulated 139.70°, theory 139.70°, Δv = 14.27 km/s
  r_p =   5 RJ: bend simulated 118.59°, theory 118.60°, Δv = 13.07 km/s
  r_p =  10 RJ: bend simulated  97.90°, theory  97.91°, Δv = 11.46 km/s
  r_p =  20 RJ: bend simulated  74.49°, theory  74.51°, Δv =  9.20 km/s
  r_p =  40 RJ: bend simulated  51.42°, theory  51.45°, Δv =  6.59 km/s
  r_p =  80 RJ: bend simulated  32.14°, theory  32.19°, Δv =  4.21 km/s
```

Table 4. Simulated flybys at v_∞ = 7.6 km/s.

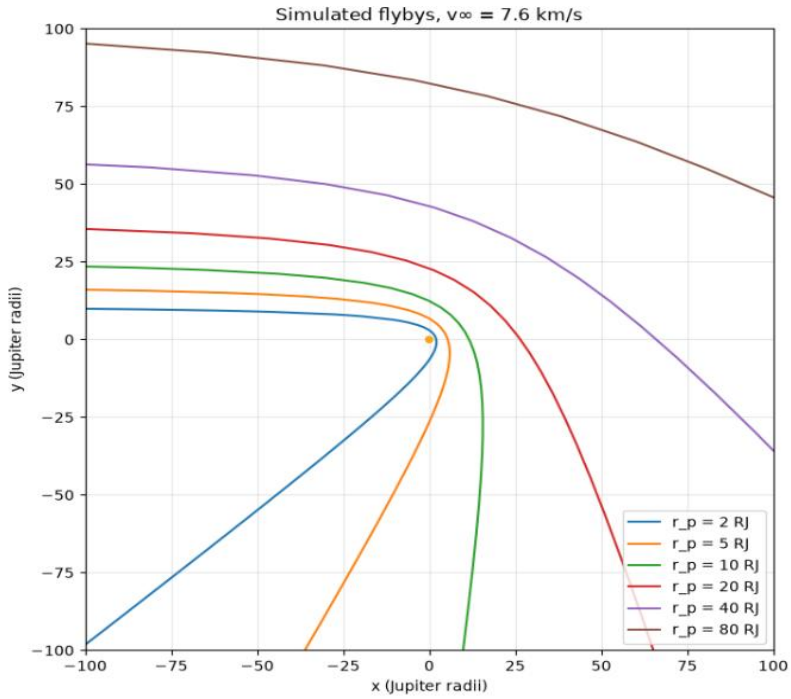


Figure 4. Simulated flyby trajectories for v∞ = 7.6 km/s at six periapsis distances.

[15]:
```
rp = np.logspace(np.log10(1.1), 2, 300) * RJ          # 1.1 to 100 RJ
fig, ax = plt.subplots(figsize=(10, 6))

for vi in [5, 7.6, 12, 18.6, 25]:
    dv = 2*vi / (1 + rp*vi**2/MU_JUP)
    ax.plot(rp/RJ, dv, label=f'v∞ = {vi} km/s')

# simulated points check the formula
for rp_RJ in [2, 5, 10, 20, 40, 80]:
    _, _, dv = simulate(rp_RJ*RJ, 7.6)
    ax.plot(rp_RJ, dv, 'kx', ms=8)

# maximum possible Δv at each distance
ax.plot(rp/RJ, np.sqrt(MU_JUP/rp), 'k--', lw=1, label='Maximum possible: √(μ/r_p)')

# the real spacecraft
for n, label in names.items():
    ax.plot(results[n]['r_p']/RJ, results[n]['dv_obs'], 'o', ms=10,
            mec='k', label=f'{label} (observed)')

ax.set_xscale('log'); ax.set_xlabel('Closest approach r_p (Jupiter radii)')
ax.set_ylabel('|Δv| (km/s)'); ax.grid(alpha=0.3, which='both')
ax.set_title('Velocity change vs closest-approach distance'); ax.legend()
plt.savefig('plots/5_dv_vs_rp.png', dpi=200); plt.show()
```

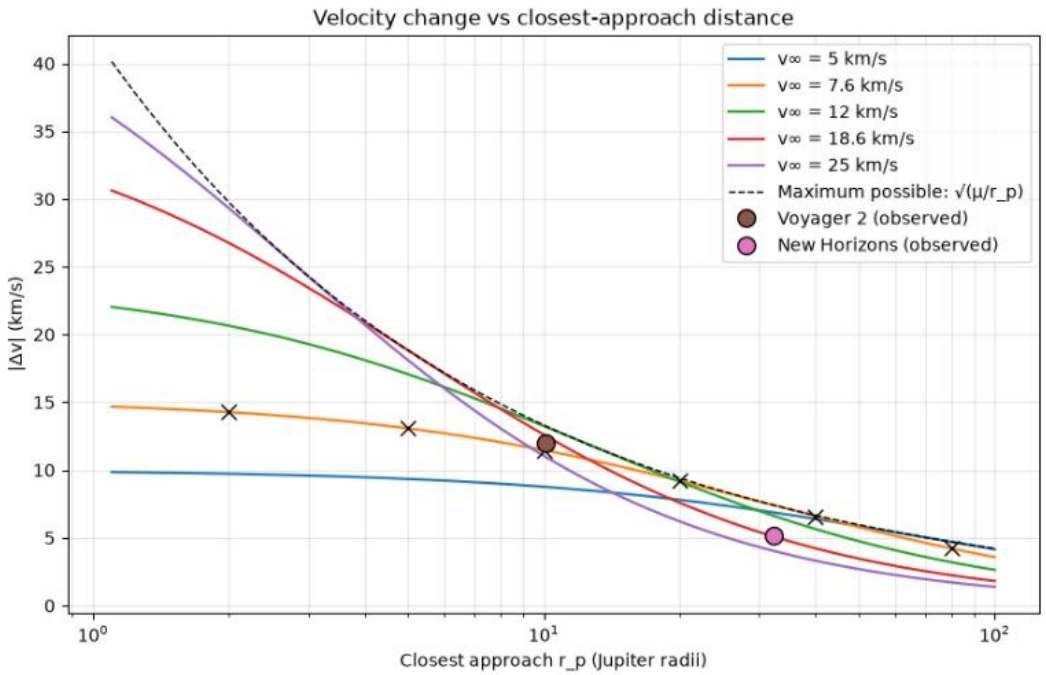


Figure 5. Analytic |Δv| against periapsis distance for five values of v∞, with the simulated points (×), the maximum possible velocity change $\sqrt{(\mu_J / r_p)}$ (dashed) and the observed values for Voyager 2 and New Horizons.

5. Fuel Savings

Table 5 gives the propellant that a single propulsive manoeuvre would require to reproduce each flyby. Using the corrected speed gain as the minimum requirement and the spacecraft's own hydrazine thrusters (I_{sp} = 220 s), Voyager 2 would need about 97 t of propellant, roughly 135 times its own mass, so that 99.3% of the departing vehicle would be fuel. Even an idealised hydrogen–oxygen engine (I_{sp} = 450 s) would require over 7 t. New Horizons would need about 1.6 t with hydrazine, 3.4 times its launch mass.

[12]:
```
G0 = 9.80665
mass = {'voyager2': 722, 'newhorizons': 478}   # spacecraft launch mass, kg
isps = [220, 320, 450]

rows = []
for n, label in names.items():
    for dv_name, dv in [('Speed gain', results[n]['speed_gain']),
                        ('|Δv| vector', results[n]['dv_obs'])]:
        for isp in isps:
            ve = isp * G0 / 1000                        # km/s
            ratio = np.exp(dv / ve)                     # m0 / mf
            fuel = mass[n] * (ratio - 1)
            rows.append({'Spacecraft': label, 'Δv type': dv_name,
                         'Δv (km/s)': round(dv, 2), 'Isp (s)': isp,
                         'Mass ratio m0/mf': round(ratio, 1),
                         'Fuel fraction (%)': round(100*(1 - 1/ratio), 1),
                         'Fuel needed (kg)': round(fuel)})

fuel_table = pd.DataFrame(rows)
fuel_table.to_csv('data/fuel_estimate.csv', index=False)
fuel_table
```

[12]:

	Spacecraft	Δv type	Δv (km/s)	Isp (s)	Mass ratio m0/mf	Fuel fraction (%)	Fuel needed (kg)
0	Voyager 2	Speed gain	10.60	220	135.9	99.3	97406
1	Voyager 2	Speed gain	10.60	320	29.3	96.6	20420
2	Voyager 2	Speed gain	10.60	450	11.0	90.9	7248
3	Voyager 2	|Δv| vector	11.95	220	254.5	99.6	183009
4	Voyager 2	|Δv| vector	11.95	320	45.1	97.8	31818
5	Voyager 2	|Δv| vector	11.95	450	15.0	93.3	10108
6	New Horizons	Speed gain	3.20	220	4.4	77.3	1628
7	New Horizons	Speed gain	3.20	320	2.8	63.9	847
8	New Horizons	Speed gain	3.20	450	2.1	51.6	509
9	New Horizons	|Δv| vector	5.14	220	10.8	90.8	4693
10	New Horizons	|Δv| vector	5.14	320	5.1	80.5	1979
11	New Horizons	|Δv| vector	5.14	450	3.2	68.8	1053

Table 5. Propellant required to match the flyby with a single burn.

The contrast between the two missions highlights the exponential character of the rocket equation. Voyager 2's speed gain is about 3.3 times that of New Horizons, yet its propellant requirement at the same specific impulse is roughly 60 times larger. A gravity assist is therefore most valuable precisely where the required velocity change is largest.

6. Discussion

6.1 Sensitivity to the measurement window

To quantify the dependence on the chosen end points, the analysis of Section 3.2 was repeated for windows of ±30, ±45 and ±60 days (Table 6). Half the spread across windows is adopted as the measurement uncertainty, giving a speed gain of 10.60 ± 0.06 km/s and |Δv| = 11.95 ± 0.23 km/s for Voyager 2, and 3.20 ± 0.03 km/s and 5.14 ± 0.01 km/s for New Horizons.

[14]:

```
rows = []
for n, label in names.items():
    s = data[n]['sun']
    for D in [30, 45, 60]:
        b = s.loc[(s['days'] + D).abs().idxmin()]
        a = s.loc[(s['days'] - D).abs().idxmin()]
        r_b = b[['x','y','z']].to_numpy(float)
        v_b = b[['vx','vy','vz']].to_numpy(float)
        v_a = a[['vx','vy','vz']].to_numpy(float)
        T = (a.jd - b.jd) * 86400
        sol = solve_ivp(sun_only, [0, T], np.concatenate([r_b, v_b]),
                        method='DOP853', rtol=1e-11, atol=1e-6)
        v_nojup = sol.y[3:, -1]
        rows.append({'Spacecraft': label, 'Window (± days)': D,
                     'Speed gain (km/s)': np.linalg.norm(v_a) - np.linalg.norm(v_nojup),
                     '|Δv| observed (km/s)': np.linalg.norm(v_a - v_nojup)})

window = pd.DataFrame(rows).round(2)
window.to_csv('data/window_sensitivity.csv', index=False)
window
```

[14]:

	Spacecraft	Window (± days)	Speed gain (km/s)	|Δv| observed (km/s)
0	Voyager 2	30	10.71	12.41
1	Voyager 2	45	10.62	12.12
2	Voyager 2	60	10.60	11.95
3	New Horizons	30	3.15	5.16
4	New Horizons	45	3.18	5.15
5	New Horizons	60	3.20	5.14

Table 6. Measured quantities as a function of window length.

The New Horizons result is insensitive to the window, as expected for end points lying about twice the SOI radius from Jupiter. For Voyager 2, by contrast, $|\Delta v|$ decreases monotonically toward the analytic value of 11.46 km/s as the window lengthens. Its trajectory is still being bent by Jupiter at ±30 and ±45 days, and even at ±60 days the spacecraft lies 44–46 × 10^6 km from Jupiter, just inside the 49 × 10^6 km SOI.

6.2 Origin of the Voyager 2 discrepancy

The 4.3% discrepancy for Voyager 2, compared with 0.6% for New Horizons, is consistent with the assumptions of the patched-conic model being less well satisfied. Voyager 2 approached at only 7.6 km/s relative to Jupiter, so it spent considerably longer in the encounter region, during which the Sun's differential pull is not negligible. The small asymmetry between its incoming and outgoing v_∞ (7.67 versus 7.59 km/s) is a direct indication of this perturbation, since a pure two-body encounter would conserve v_∞ exactly. The window trend in Table 6 suggests that a longer baseline would reduce the discrepancy further.

The New Horizons speed gain of 3.20 km/s is lower than the figure of about 4 km/s often quoted for this flyby. The difference reflects the definition of "gain", which depends on the epochs compared and, on the baseline, adopted, rather than a conflict with the data.

6.3 Limitations

Several limitations should be noted. First, the Horizons trajectories are not raw measurements, but reconstructions fitted to radio-tracking data with dynamical models that already include Jupiter's gravity. The analysis therefore tests the internal consistency of the reconstructed trajectories with gravity-assist physics rather than providing an independent measurement. Second, the patched-conic model neglects the Galilean satellites, Jupiter's oblateness and any trajectory correction manoeuvres near the encounter. Third, the Sun-only baseline is itself a model. Finally, the fuel estimate assumes a single impulsive burn with no staging and ignores the additional tank mass and launch-vehicle capacity that the extra propellant would demand; it is therefore best regarded as a lower bound on the real cost.

7. Conclusion

The velocity gained by a spacecraft during a planetary flyby can be measured directly from publicly available trajectory data, and the result agrees closely with gravity-assist theory. Jupiter increased the heliocentric speed of Voyager 2 by 10.60 ± 0.06 km/s, more than doubling it, and that of New Horizons by 3.20 ± 0.03 km/s. The patched-conic model reproduces the measured velocity changes to within 4.3% and 0.6% and the deflection angles to within 0.6°, while an independent numerical integration reproduces the analytic deflections to within 0.05°. In Jupiter's own frame the spacecraft departed with essentially the speed at which it arrived, confirming that the flyby redirects rather than accelerates the spacecraft locally. Achieving the same gains propulsively

would have required tens of tonnes of propellant for Voyager 2, well beyond what any spacecraft of its size could carry.

Future work could extend the Voyager 2 window to ±120 days to test whether the residual discrepancy disappears, apply the same method to the Saturn, Uranus and Neptune encounters, and incorporate solar perturbations during the encounter itself.

References

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